

Practical Workbook
CT-361
Artificial Intelligence & Expert Systems



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Q1. Write a Prolog program to find the length of a list.

SOLUTION:

```
%Q1
list_length([], 0).
list_length(_|T, Length) :-
    list_length(T, N),
    Length is N+1.
```

OUTPUT:

```
105 ?- list_length([apple,mango,banana],Length).
Length = 3.
```

Q2. Write a Prolog program to check whether a given element exists in a list.

SOLUTION:

```
%Q2
element_exists(X, [X|_]).

element_exists(X, [_|T]) :-
    element_exists(X, T).
```

OUTPUT:

```
108 ?- element_exists(apple, [apple, mango, banana]).
true .
```

Q3. Write a Prolog program to find the sum of all elements in a numeric list.

SOLUTION:

```
%Q3
list_sum([],0).
list_sum([H|T],Sum):-
    list_sum(T,N),
    Sum is H+N.
```

OUTPUT:

```
112 ?- list_sum([1,2,3,4,5],Sum).
Sum = 15.
```

Q4. Write a Prolog program to find the largest element in a list.

SOLUTION:

```
%Q4
largest([X],X).
largest([H|T],Max):-
    largest(T,N),
    (H>N->Max=H ;Max=N).
```

OUTPUT:

```
115 ?- largest([41,73,24,90,50],Max).
Max = 90 .
```

Q5. Write a Prolog program to reverse a list.

SOLUTION:

```
%Q5
reversethelist([],[]).
reversethelist([H|T],R):-
    reversethelist(T,N),
    append(N,[H],R).
```

OUTPUT:

```
117 ?- reversethelist([apple,mango,banana],R).  
R = [banana, mango, apple].
```

Q6. Write a Prolog program to count the number of even elements in a list.

SOLUTION:

```
%Q6  
count_even([],0).  
count_even([H|T],Count):-  
    count_even(T,N),  
    (H mod 2:=0->Count is N+1;Count=N).
```

OUTPUT:

```
119 ?- count_even([1,2,4,7,8],Count).  
Count = 3.
```

Q7. Write a Prolog program to remove a specific element from a list.

SOLUTION:

```
%Q7  
remove(X,[X|T],T).  
remove(X,[H|T],[H|R]):-  
    remove(X,T,R).
```

OUTPUT:

```
122 ?- remove(mango,[apple,mango,banana],R).  
R = [apple, banana] .
```